Programming Methodologies
Section 5 and 13 Dr. Abdul Raouf Khan

EXERCISE NO 1.

Submitted By or Before: 13 September 2026

1. Both main memory and secondary storage are types of memory. Describe the difference between the two.
   Answer: Main memory is responsible for holding the currently-working programs, that memory is the RAM. It holds the data until the computer is turned off. The secondary memory is for long-term storing, such as SSD. Unlike the main memory, the secondary memory holds the data even when the The computer is not working.

2. What is the difference between system software and application software?
   Answer: The system software is the programs that's responsible for running the computer (aka the operating system). On the other hand, the application software is a program that performs a certain task for the user. Like the MS Word, Excel, Calculator App, etc.

3. What type of software controls the internal operations of the computer's hardware?
   Answer: System software.

4. Why must programs written in a high-level language be translated into machine language before they can be run?
   Answer: As they can't be read by the computer. The computer only understands the machine language.

5. Why is it easier to write a program in a high-level language than in machine language?
   Answer: As it is much more readable than machine language.

6. Explain the difference between an object file and an executable file.
   Answer: The object file is the compiled source code file of a program. While the executable file is the Ready-to-run program file, it is produced by the linker after it links the object file with the Needed header files.

7. What is the difference between a syntax error and a logical error?
   Answer: The core difference between a syntax and logical error is that the syntax error will stop The compiler from building the executable. In contrast, the logical error won't prevent the compiler From finishing its task, leading to an unwanted output and wrong functionality.

Fill-in-the-Blank

8. Computers can do many different jobs because they can be **programmed**.
9. The job of the **CPU** is to fetch instructions, carry out the operations commanded by the instructions, and produce some outcome or resultant information.
10. Internally, the CPU consists of the **Arithmetic and Logic Unit** and the **Control Unit**.
11. A(n) **SSD** is an example of a secondary storage device.
12. The two general categories of software are **System Software** and **Application Software**.
13. A program is a set of **instructions**.
14. Since computers can't be programmed in natural human language, algorithms must be written in a(n) **Programming** language.
15. **Machine Language** is the only language computers really process.
16. **High-Level** languages are close to the level of humans in terms of readability.
17. **Low-Level** languages are close to the level of the computer.
18. A program's ability to run on several different types of computer systems is called **Portable**.
19. Words that have special meaning in a programming language are called **Keywords**.
20. Words or names defined by the programmer are called **Identifiers**.
21. **Operators** are characters or symbols that perform operations on one or more operands.
22. **Punctuation** characters or symbols mark the beginning or end of programming statements, or separate items in a list.
23. The rules that must be followed when constructing a program are called **Syntax**.
24. A(n) **Variable** is a named storage location.
25. A variable must be **defined** before it can be used in a program.
26. The three primary activities of a program are **Input** , **Processing**, and **Output**.
27. **Input** is information a program gathers from the outside world.
28. **Output** is information a program sends to the outside world.
29. A(n) **Flowchart** is a diagram that graphically illustrates the structure of a program.

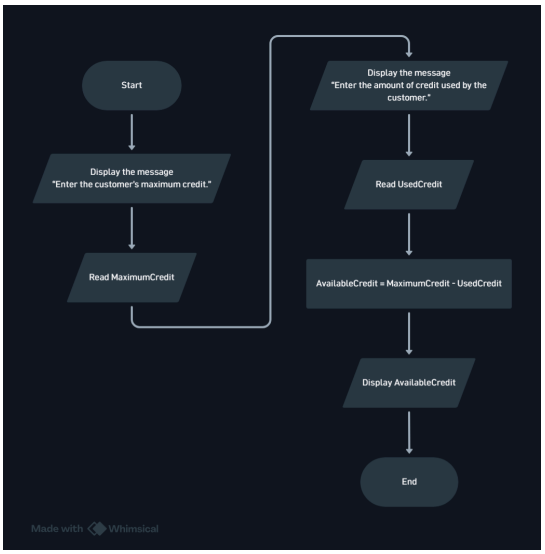
Algorithm Workbench

Draw hierarchy charts or flowcharts that depict the programs described below. (See Appendix C for instructions on creating flowcharts.)

30. Available Credit

The following steps should be followed in a program that calculates a customer's available credit:

1. Display the message "Enter the customer's maximum credit."
2. Wait for the user to enter the customer's maximum credit.
3. Display the message "Enter the amount of credit used by the customer."
4. Wait for the user to enter the customer's credit used.
5. Subtract the used credit from the maximum credit to get the customer's available credit.
6. Display a message that shows the customer's available credit.



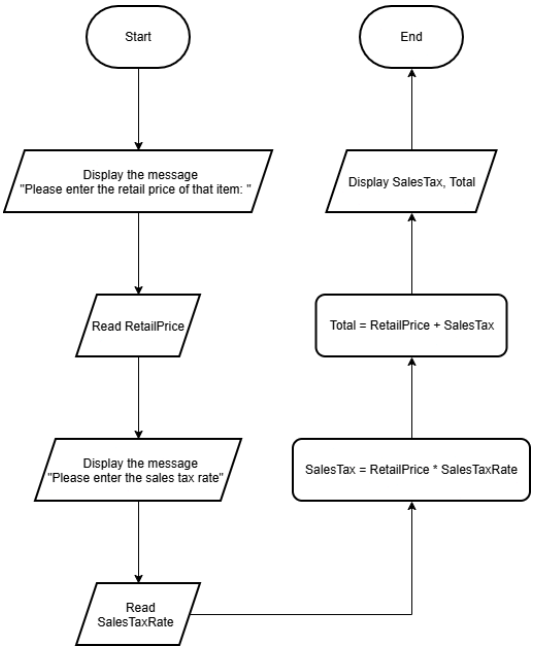
31. Sales Tax

Design a hierarchy chart or flowchart for a program that calculates the total of a retail sale. The program should ask the user for:

– The retail price of the item being purchased
– The sales tax rate

Once these items have been entered, the program should calculate and display:

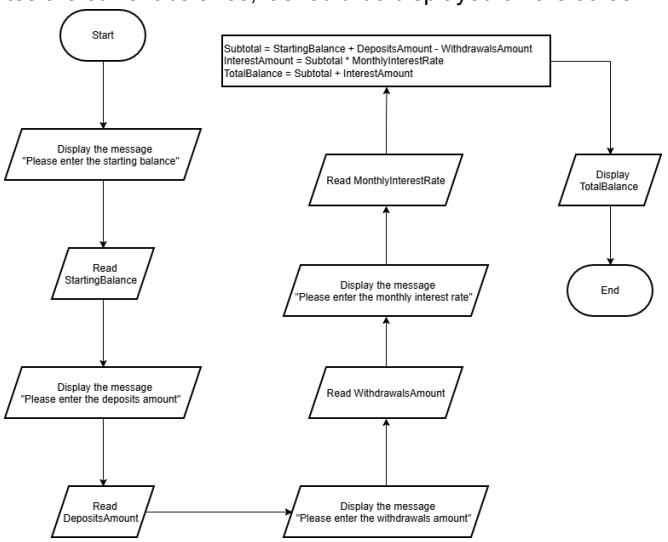
– The sales tax for the purchase
– The total of the sale



32. Account Balance
Design a hierarchy chart or flowchart for a program that calculates the current balance in a savings account. The program must ask the user for:

– The starting balance
– The total dollar amount of deposits made
– The total dollar amount of withdrawals made
– The monthly interest rate

Once the program calculates the current balance, it should be displayed on the screen.



Predict the Result

Questions 33–35 are programs expressed as English statements. What would each display on the screen if they were actual programs?

33. The variable x starts with the value 0.
The variable y starts with the value 5.
Add 1 to x.
Add 1 to y.
Add x and y, and store the result in y.
Display the value in y on the screen.
Answer: y=7

34. The variable j starts with the value 10.
The variable k starts with the value 2.
The variable l starts with the value 4.
Store the value of j times k in j. j=20
Store the value of k times l in l. l=8
Add j and l, and store the result in k. k=160
Display the value in k on the screen.
Answer: k=160

35. The variable a starts with the value 1.
The variable b starts with the value 10.
The variable c starts with the value 100.
The variable x starts with the value 0.
Store the value of c times 3 in x. x=100*3=300
Add the value of b times 6 to the value already in x. x=x+10*6=360
Add the value of a times 5 to the value already in x. x=x+1*5 = 365
Display the value in x on the screen. x=365

Find the Errors

36. The following *pseudocode algorithm* has an error. The program is supposed to ask the user for the length and width of a rectangular room, then display the room's area. The program must multiply the width by the length in order to determine the area. Find the error.

area = width × length. (width and height are not defined or don't have values yet. That's the error).
Display "What is the room's width?".
Input width.
Display "What is the room's length?".
Input length.
Display area.